👩‍🏫 Staff Guide · Ag Science 1 · Unit 2 Washington Crops: Potatoes, Wheat & Apples (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Mr. Torretta's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Bedlington Farm PowerPoint, WA Wheat Barley and Potatoes, Potato Lab, Cosmic Crisp Apples, Apple Taste Lab (Ms. Gholam)”)	**Level 2 piece**
Bedlington Farm PowerPoint (Blank Out, 25 pts)	Notes A–B + slides; exact blank answers (number of growers in 1942, acreage in 1997, overseas shipping): check with Mr. Torretta
WA Wheat, Barley and Potatoes (12 pts) · Cosmic Crisp (10 pts)	Video guides
Potato Lab (30 pts)	Potato Lab packet (his lab sheet: check with Mr. Torretta)
Apple Taste Lab (Ms. Gholam, 20 pts)	Apple Taste Lab packet
Class slides	Deck: 20 slides, Words to Know first, hold-up-1-2-3 checks, recap
Class notes (Blank Out)	Guided notes: 17 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	10 picture cards (cut apart)
—	Review game (match + order) on the class site

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Real potatoes: find the eyes
- Taste or look at 2 apple varieties and sort sweet vs. tart

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Video guides: L1 picture choices, L2 3 choices + starters; same points
- **Notes**: same points as the class Blank Out notes when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with Mr. Torretta”.

6. Photo credits (open licenses)

Wheat harvest.jpg — Unknown authorUnknown author, Public domain, Wikimedia Commons · Barley in Slovenia.jpg — Lucash, CC BY-SA 3.0, Wikimedia Commons · Potato Harvest - geograph.org.uk - 7256460.jpg — KS, CC BY-SA 2.0, Wikimedia Commons · Potato Flower on the field.jpg — Ezhuttukari, CC BY-SA 3.0, Wikimedia Commons · Potato sprout, January 23, 2006.jpg — unknown, CC BY-SA 2.0, Wikimedia Commons · Cosmic Crisp apple in kitchen.jpg — Bri, CC BY-SA 4.0, Wikimedia Commons · Cosmic Crisp (apple) jm141839.jpg — joergens.mi, CC BY-SA 3.0, Wikimedia Commons · Honey bee on apple blossom, Sandy, Bedfordshire (6957165968).jpg — Orangeaurochs from Sandy, Bedfordshire, United Kingdom, CC BY 2.0, Wikimedia Commons · Diagrams/pictures from Mr. Torretta's class slides: potato-diagram.png, potato-plant.jpg, potato-planting.jpg, potato-purple.jpg, potato-russet.jpg, sprout.png

👩🏾‍🏫 Staff Guide · Apple Taste Lab (Level 2)

Ag Science 1 · Unit 2 · goes with the class “Apple Taste Lab (Ms. Gholam)” (20 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: Apples with Ms. Gholam: pre-lab questions, video (youtube.com/watch?v=UWLmEh1HIBw), and a tasting chart for 6 varieties. 6 steps, one per box. **check with Mr. Torretta**: whether the lab is still done this year and the allergy rules
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Sweet or sour?”
- “Point to the face that shows how you liked it.”
- “Which one is your favorite?”

3. Modifications

- Chart with a word bank
- 5 pre-lab/video questions instead of 15

4. What to grade (20 pts)

- Same 20 points: chart + questions
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Golden Delicious, SweeTango, Granny Smith, and Gala apples 3.JPG — Dllu, CC BY-SA 4.0, Wikimedia Commons

👩‍🏫 Staff Guide · Potato Lab (Level 2)

Ag Science 1 · Unit 2 · goes with the class “Potato Lab” (30 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: Potato Lab (30 pts). The export page is blank, so this packet is a potato variety observation lab built on the Bedlington Farms potato varieties. 7 steps, one per box. **check with Mr. Torretta**: his actual Potato Lab sheet and whether it includes cooking
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Point to an eye.”
- “What color is the inside?”
- “Which one is the biggest?”

3. Modifications

- Chart with 6 columns and a word bank on the board
- 3 short questions

4. What to grade (30 pts)

- Same 30 points: chart + 3 questions
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Diagrams/pictures from Mr. Torretta's class slides: potato-russet.jpg

🥔 Washington Crops: Potatoes, Wheat and Apples

Ag Science 1 · Unit 2 Washington Crops: Potatoes, Wheat & Apples · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Dick Bedlington Farms

1. The farm started in _____ with 4 acres.
2. Whatcom County is an _____ area, so there are fewer insects and diseases.
3. They make clean seed with tissue _____ in a lab.
4. In the greenhouse they harvest _____.
5. They sell seed potatoes to commercial potato _____.

Word bank: **1942** **culture** **growers** **isolated** **minitubers**

B. Potato plant and wheat

6. Match the potato plant part:

Eye	make food from sunlight
Tuber	blooms on top
Leaves	where a sprout grows
Flower	the potato we eat

7. Washington mostly grows soft _____ wheat.
8. Most Washington wheat is _____ to other countries.
9. Columbia Basin farms _____ because there is little rain.

Word bank: **exported** **irrigate** **white**

C. Apples

10. Washington's top crop is _____.
11. _____ pollinate apple blossoms.
12. Cosmic Crisp is a cross of Enterprise and _____.
13. Cosmic Crisp was developed at _____.
14. Circle 2 reasons growers like Cosmic Crisp:

stores a long time | always green | crisp and juicy | grows underground

Word bank: **apples** **Bees** **Honeycrisp** **WSU**

Level 2 version of the gen ed "Dick Bedlington Farms (Blank Out) + Wheat & Potatoes video + Cosmic Crisp + Apple Lab."

📖 Washington Crops: Potatoes, Wheat & Apples: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



🥔

Potato

A tuber: the swollen underground stem of the potato plant.



🌱

Seed potato

A potato (or piece) planted to grow a new potato plant. Bedlington Farms grows certified seed potatoes.



👁️

Eye

The bud on a potato where a new sprout grows.

🧺

Harvest

Gathering a crop when it is ready. Potato vines dying back is a sign it is time.



Wheat

A grain crop for flour. Most Washington wheat is soft white wheat, and most is exported.



Barley

A grain used for animal feed and malt.



Variety

One kind of a crop, like Gala, Fuji, or Cosmic Crisp apples.

Orchard

A field of fruit trees, like apples or cherries.

Grafting

Joining a branch of one tree onto the roots of another, so the new tree grows the same apple.



Cosmic Crisp

An apple bred by WSU from Enterprise × Honeycrisp. It stores for a long time.

🍎 Apple Taste Lab

Ag Science 1 · Unit 2 · Level 2



Name: _____

Apples with Ms. Gholam: pre-lab questions, video (youtube.com/watch?v=UWLmEh1HIBw), and a tasting chart for 6 varieties · check with Mr. Torretta.

🧺 6 apple varieties (slices)

- ☐ 🍎 **4** Gala slices
- ☐ 🍏 **4** Granny Smith slices
- ☐ 🍎 **4** Envy slices
- ☐ 🍎 **4** Fuji slices
- ☐ 🍯 **4** Honeycrisp slices
- ☐ 🌌 **4** Cosmic Crisp slices

🌱 Steps: check each box when you finish

1. ❓ Answer the pre-lab questions. ☐
2. 🎬 Watch the video and answer the video questions. ☐
3. 🧼 Wash hands; check allergies. ☐
4. 🍎 Taste each variety; drink water between. ☐
5. 📝 Record taste, texture, and best use. ☐
6. 🏆 Rank your top 3 and explain your favorite. ☐

🍎 Apple taste chart

Variety	Taste	Texture	Use
🍎 Gala			
🍏 Granny Smith			
🍎 Envy			
🍎 Fuji			
🍯 Honeycrisp			
🌌 Cosmic Crisp			

Word bank: **sweet** **tart** **crisp** **juicy** **soft** **snack** **baking** **salad**

Pre-lab and video

1. Apples came from the mountains of _____ (DNA). *Kazakhstan*
2. Apples are a good source of _____ and _____.
3. Can an apple seed grow the exact same apple? Yes / No
4. What is grafting? _____
5. What insect is brought in when the trees bloom? _____
6. My top 3: 1. _____ 2. _____ 3. _____

💡 *Apple seeds don't grow the same apple. Growers graft branches to keep each variety true.*

🥔 Potato Lab

Ag Science 1 · Unit 2 · Level 2



Name: _____

Potato Lab (30 pts). The export page is blank, so this packet is a potato variety observation lab built on the Bedlington Farms potato varieties · check with Mr. Torretta.

🧺 Per group

- ☐ 🟤 **1** russet potato
- ☐ 🔴 **1** red potato
- ☐ 🟡 **1** yellow potato
- ☐ 🟣 **1** purple potato
- ☐ 👆 **1** fingerling potato

🌱 Steps: check each box when you finish

1.  Wash the 5 potatoes. ☐
2.  Measure each potato's length (cm). ☐
3.  Count the eyes on each one. ☐
4.  Cut each in half (adult supervising). ☐
5.  Record skin color, flesh color, and texture. ☐
6.  Predict the best use: bake, mash, roast, or salad. ☐
7.  Clean up and answer the questions. ☐

🥔 Potato variety chart

Variety	Length (cm)	Eyes	Skin color	Flesh color	Texture	Best use
🟤 Russet						
🔴 Red						
🟡 Yellow						
🟣 Purple						
👆 Fingerling						

Questions

1. Which potato had the most eyes? _____
2. What grows from a potato's eye? _____
3. Bedlington Farms grows seed potatoes. What is a seed potato?

_____

💡 *Starchy russets are best for baking and fries; waxy reds and fingerlings hold their shape in salads and roasting.*

🍎 Apple Taste Lab · Prep Sheet

Ag Science 1 · Unit 2 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🏷️ **Labeler**	Labels the plates with the apple names.	
🍽️ **Server**	Passes the plates.	
📝 **Recorder**	Fills in the group chart.	
🧽 **Cleaner**	Cleans the table.	

🧰 Tools and supplies: check when you have it

- ☐ 🍽️ **Plates** one per variety, labeled
- ☐ 🔪 **Knife + board** staff slices
- ☐ 🧻 **Napkins** one each
- ☐ 💧 **Water** to clear your taste

⚠️ Safety and manners focus

⚠️ Allergy check before tasting. You can look and smell instead.

🧼 Wash hands first.

🔪 Staff does the slicing.

🥔 Potato Lab · Prep Sheet

Ag Science 1 · Unit 2 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🧼 **Washer**	Washes the potatoes.	
🔪 **Cutter**	Cuts each potato in half (with an adult).	
🔍 **Looker**	Counts eyes and checks colors.	
📝 **Recorder**	Fills in the chart.	

🧰 Tools and supplies: check when you have it

- ☐ 🔪 **Knife** adult or trained student only
- ☐ 🪵 **Cutting board** one per group
- ☐ 🔍 **Hand lens** to see eyes and skin
- ☐ 📏 **Ruler** measure length

⚠️ Safety and manners focus

🔪 Knives: claw grip, cut away from you, adult watching.

🧼 Wash hands and potatoes first.

🚫 Do not eat raw potato or green skin.

🍎 Apple Taste Lab · Reflection

Ag Science 1 · Unit 2 · Level 2



Name: _____

📸 My photos / proof

☐ Plates labeled ☐ Chart done

✅ Skills I used today (check all you did)

☐ 🍎 I tested all 6 varieties

☐ 📝 I filled in taste, texture, and use

☐ 🏆 I ranked my top 3

☐ 🧼 I followed the allergy and hand rules

⭐ Which apple would you buy? Why?

1. _____
2. _____

Ideas: **sweet** **crisp** **good for baking**

🔧 Which surprised you?

1. _____
2. _____

Ideas: **Granny Smith** **Cosmic Crisp** **Envy**

🥔 Potato Lab · Reflection

Ag Science 1 · Unit 2 · Level 2



Name: _____

📸 My photos / proof

☐ Potatoes cut ☐ Chart done ☐ Clean station

✅ Skills I used today (check all you did)

☐ 👀 I recorded 5 varieties

☐ 👁️ I counted eyes and measured

☐ 🔪 I used knife safety

☐ 🤝 I did my role

⭐ Which potato would you grow? Why?

1. _____
2. _____

Ideas: **color** **taste** **best use**

🔧 What surprised you?

1. _____
2. _____

Ideas: **purple inside** **number of eyes** **size**

✅ Washington Crops: Potatoes, Wheat & Apples Quiz

Ag Science 1 · Unit 2 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🥔 Bedlington Farms grows…

A. Apples | B. Certified seed potatoes | C. Dairy cows

2. 🛡️ Whatcom is good for seed because it is…

A. Very hot | B. In the desert | C. Isolated, with fewer pests

3. 🧪 Clean seed starts with…

A. Tissue culture in a lab | B. Store potatoes | C. Wheat seeds

4. 👁️ A potato's eyes…

A. See light | B. Hold water | C. Sprout new stems

5. 🌾 Washington mostly grows…

A. Soft white wheat | B. Rice | C. Cotton

6. 🚢 Most Washington wheat is…

A. Thrown away | B. Exported | C. Fed to fish

7. 💧 Columbia Basin potato farms…

A. Get lots of rain | B. Irrigate from the river | C. Grow in snow

8. 🧺 A sign potatoes are ready:

A. The vines die back | B. Flowers open | C. Leaves turn blue

9. 🌌 Cosmic Crisp was bred at…

A. Disney | B. NASA | C. WSU

10. 🍎 Cosmic Crisp's parents are Enterprise and…

A. Honeycrisp | B. Fuji | C. Gala

🎬 Cosmic Crisp Apples

Ag Science 1 · Unit 2 · Level 2 video guide (modified). YouTube: youtube.com/watch?v=Uzg9MjlVTtY

Name: _____

Before you watch

What is your favorite apple? _____

While you watch: circle the best answer

1. Enterprise apples were crossed with what apple?

A. Granny Smith | B. Honeycrisp | C. Red Delicious

2. Who invented Cosmic Crisp? Dr. ___ Barritt

A. Bob | B. Ben | C. Bruce

3. Who owns the patent?

A. Washington State University | B. Apple Inc. | C. A grocery store

4. A built-in advantage of Cosmic Crisp:

A. It never needs water | B. It grows in snow | C. It stores a long time and stays crisp

5. The industry taxed itself to create Washington Tree Fruit…

A. Research Commission | B. Stores | C. Bakeries

After you watch: finish the sentence

6. Why was Cosmic Crisp developed?

_____ _____

7. Have you tried it? What did you think?

_____ _____

Goes with the class “Cosmic Crisp Apples” (10 pts). Same points when complete.

🎬 Washington Wheat, Barley and Potatoes

Ag Science 1 · Unit 2 · Level 2 video guide (modified). 3 YouTube videos (youtube.com/watch?v=…): YEkSO4E-jwg wheat · g5TwHUA1NnQ potatoes · _SK75rl2Yao growing

Name: _____

While you watch: circle the best answer

1. What does Washington mostly produce?

A. Rice | B. Soft white wheat | C. Cotton

2. Most Washington wheat is…

A. Fed to fish | B. Thrown away | C. Exported to other countries

3. One of the best places to grow potatoes is…

A. The Columbia Basin | B. The North Pole | C. A rain forest

4. Columbia Basin farms need…

A. Snow plows | B. Boats | C. Irrigation (little rain)

5. A new potato plant grows from…

A. A seed potato (its eyes) | B. A potato leaf | C. The flower

6. A sign it is time for the potato harvest:

A. Leaves turn blue | B. The vines die back | C. It snows

After you watch: finish the sentence

7. Wheat that falls over is called

_____ _____

Goes with the class “WA Wheat, Barley and Potatoes” (12 pts). Same points when complete.

🥔 Washington Crops: Potatoes, Wheat and Apples · ANSWER KEY

Ag Science 1 · Unit 2 Washington Crops: Potatoes, Wheat & Apples · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Dick Bedlington Farms

1. The farm started in **1942** with 4 acres.
2. Whatcom County is an **isolated** area, so there are fewer insects and diseases.
3. They make clean seed with tissue **culture** in a lab.
4. In the greenhouse they harvest **minitubers**.
5. They sell seed potatoes to commercial potato **growers**.

Word bank: 1942 · culture · growers · isolated · minitubers

B. Potato plant and wheat

6. Match the potato plant part:

(Key: Eye → where a sprout grows · Tuber → the potato we eat · Leaves → make food from sunlight · Flower → blooms on top)

Eye	make food from sunlight
Tuber	blooms on top
Leaves	where a sprout grows
Flower	the potato we eat

7. Washington mostly grows soft **white** wheat.
8. Most Washington wheat is **exported** to other countries.
9. Columbia Basin farms **irrigate** because there is little rain.

Word bank: exported · irrigate · white

C. Apples

10. Washington's top crop is **apples**.
11. **Bees** pollinate apple blossoms.
12. Cosmic Crisp is a cross of Enterprise and **Honeycrisp**.
13. Cosmic Crisp was developed at **WSU**.
14. Circle 2 reasons growers like Cosmic Crisp:

stores a long time | always green | **crisp and juicy** | grows underground

Word bank: **apples** **Bees** **Honeycrisp** **WSU**

Level 2 version of the gen ed "Dick Bedlington Farms (Blank Out) + Wheat & Potatoes video + Cosmic Crisp + Apple Lab."

✅ Washington Crops: Potatoes, Wheat & Apples Quiz · ANSWER KEY

Ag Science 1 · Unit 2 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🥔 Bedlington Farms grows… Bedlington

A. Apples | **B. Certified seed potatoes** | C. Dairy cows

2. 🛡️ Whatcom is good for seed because it is… Bedlington

A. Very hot | B. In the desert | **C. Isolated, with fewer pests**

3. 🧪 Clean seed starts with… Bedlington

A. Tissue culture in a lab | B. Store potatoes | C. Wheat seeds

4. 👁️ A potato's eyes… Bedlington

A. See light | B. Hold water | **C. Sprout new stems**

5. 🌾 Washington mostly grows… Wheat video

A. Soft white wheat | B. Rice | C. Cotton

6. 🚢 Most Washington wheat is… Wheat video

A. Thrown away | **B. Exported** | C. Fed to fish

7. 💧 Columbia Basin potato farms… Wheat video

A. Get lots of rain | **B. Irrigate from the river** | C. Grow in snow

8. 🧺 A sign potatoes are ready: Wheat video

A. The vines die back | B. Flowers open | C. Leaves turn blue

9. 🌌 Cosmic Crisp was bred at… Cosmic Crisp

A. Disney | B. NASA | **C. WSU**

10. 🍎 Cosmic Crisp's parents are Enterprise and… Cosmic Crisp

A. Honeycrisp | B. Fuji | C. Gala

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "Bedlington Farm, Wheat & Potatoes video, Cosmic Crisp". Score: __ / 10.

🎬 Cosmic Crisp Apples · ANSWER KEY

Ag Science 1 · Unit 2 · Level 2 video guide (modified). YouTube: youtube.com/watch?v=Uzg9MjlVTtY

Name: _____

Before you watch

What is your favorite apple? _____

While you watch: circle the best answer

1. Enterprise apples were crossed with what apple?

A. Granny Smith | **B. Honeycrisp** | C. Red Delicious

2. Who invented Cosmic Crisp? Dr. ___ Barritt

A. Bob | B. Ben | **C. Bruce**

3. Who owns the patent?

A. Washington State University | B. Apple Inc. | C. A grocery store

4. A built-in advantage of Cosmic Crisp:

A. It never needs water | B. It grows in snow | **C. It stores a long time and stays crisp**

5. The industry taxed itself to create Washington Tree Fruit…

A. Research Commission | B. Stores | C. Bakeries

After you watch: finish the sentence

6. Why was Cosmic Crisp developed?

_____ **a new, high-quality apple bred for Washington growers**

7. Have you tried it? What did you think?

_____ _____

Goes with the class "Cosmic Crisp Apples" (10 pts). Same points when complete. Number of trees planted when the video was made: check with Mr. Torretta.

🎬 Washington Wheat, Barley and Potatoes · ANSWER KEY

Ag Science 1 · Unit 2 · Level 2 video guide (modified). 3 YouTube videos (youtube.com/watch?v=…): YEkSO4E-jwg wheat · g5TwHUA1NnQ potatoes · _SK75rl2Yao growing

Name: _____

While you watch: circle the best answer

1. What does Washington mostly produce?

A. Rice | **B. Soft white wheat** | C. Cotton

2. Most Washington wheat is…

A. Fed to fish | B. Thrown away | **C. Exported to other countries**

3. One of the best places to grow potatoes is…

A. The Columbia Basin | B. The North Pole | C. A rain forest

4. Columbia Basin farms need…

A. Snow plows | B. Boats | **C. Irrigation (little rain)**

5. A new potato plant grows from…

A. A seed potato (its eyes) | B. A potato leaf | C. The flower

6. A sign it is time for the potato harvest:

A. Leaves turn blue | **B. The vines die back** | C. It snows

After you watch: finish the sentence

7. Wheat that falls over is called

_____ **lodging — check with Mr. Torretta**

Goes with the class "WA Wheat, Barley and Potatoes" (12 pts). Same points when complete. First wheat crop place and date, yields 100 years ago, export %, and inches of rain are video-specific; check with Mr. Torretta.